

Problem Statement

For $a \in \mathbb{R}, a \neq -1$, find a such that:

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n k^a}{(n+1)^{a-1} \sum_{k=1}^n (na+k)} = \frac{1}{60}$$

Detailed Solution

To evaluate this limit, we can manipulate the expression to express it in terms of Riemann sums, which can then be evaluated as definite integrals. Let's break this down step-by-step.

Step 1: Algebraic manipulation of the expression

First, let's analyze the numerator and the denominator by factoring out powers of n .

For the numerator, we can factor out n^a :

$$\sum_{k=1}^n k^a = \sum_{k=1}^n n^a \left(\frac{k}{n}\right)^a = n^{a+1} \cdot \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^a$$

For the denominator, we analyze the two factors separately. First, factor out n from the base of the exponent:

$$(n+1)^{a-1} = n^{a-1} \left(1 + \frac{1}{n}\right)^{a-1}$$

Next, look at the summation in the denominator. We can factor out n :

$$\sum_{k=1}^n (na+k) = \sum_{k=1}^n n \left(a + \frac{k}{n}\right) = n^2 \cdot \frac{1}{n} \sum_{k=1}^n \left(a + \frac{k}{n}\right)$$

Now, substitute these modified components back into the original fraction:

$$\frac{n^{a+1} \cdot \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^a}{n^{a-1} \left(1 + \frac{1}{n}\right)^{a-1} \cdot n^2 \cdot \frac{1}{n} \sum_{k=1}^n \left(a + \frac{k}{n}\right)}$$

Notice that the powers of n in the denominator combine to $n^{a-1} \cdot n^2 = n^{a+1}$. This perfectly cancels with the n^{a+1} in the numerator:

$$\frac{\frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^a}{\left(1 + \frac{1}{n}\right)^{a-1} \frac{1}{n} \sum_{k=1}^n \left(a + \frac{k}{n}\right)}$$

Step 2: Evaluate the limit using definite integrals

Taking the limit as $n \rightarrow \infty$, we can evaluate each part. Assuming $a > -1$, the Riemann sums converge to definite integrals over the interval $[0, 1]$:

For the numerator:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n}\right)^a = \int_0^1 x^a dx = \left[\frac{x^{a+1}}{a+1}\right]_0^1 = \frac{1}{a+1}$$

For the denominator's sum:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(a + \frac{k}{n}\right) = \int_0^1 (a+x) dx = \left[ax + \frac{x^2}{2}\right]_0^1 = a + \frac{1}{2}$$

For the remaining exponential factor in the denominator:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{a-1} = 1^{a-1} = 1$$

Combining these limits, the original limit L simplifies significantly:

$$L = \frac{\frac{1}{a+1}}{1 \cdot \left(a + \frac{1}{2}\right)} = \frac{1}{(a+1) \left(a + \frac{1}{2}\right)}$$

Step 3: Solve for a

We are given that the limit evaluates to $\frac{1}{60}$. We set our simplified expression equal to this value:

$$\frac{1}{(a+1) \left(a + \frac{1}{2}\right)} = \frac{1}{60}$$

Cross-multiply to create a polynomial:

$$(a+1) \left(a + \frac{1}{2}\right) = 60$$

$$a^2 + \frac{3}{2}a + \frac{1}{2} = 60$$

Multiply the entire equation by 2 to clear the fractions:

$$2a^2 + 3a + 1 = 120$$

$$2a^2 + 3a - 119 = 0$$

Now, apply the quadratic formula, $a = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$\begin{aligned} a &= \frac{-3 \pm \sqrt{3^2 - 4(2)(-119)}}{2(2)} \\ &= \frac{-3 \pm \sqrt{9 + 952}}{4} \\ &= \frac{-3 \pm \sqrt{961}}{4} \\ &= \frac{-3 \pm 31}{4} \end{aligned}$$

This yields two possible values for a :

$$a_1 = \frac{28}{4} = 7 \quad \text{and} \quad a_2 = \frac{-34}{4} = -8.5$$

Step 4: Verify the domain of a

In Step 2, we assumed $a > -1$ to ensure the convergence of the Riemann sum to $\int_0^1 x^a dx$. If $a = -8.5$, the integral $\int_0^1 x^{-8.5} dx$ diverges.

More rigorously, if $a < -1$, the sum in the numerator $\sum_{k=1}^n k^a$ converges to a constant (related to the Riemann zeta function). Meanwhile, the denominator evaluates asymptotically to $n^{a+1} (a + \frac{1}{2})$. Since $a + 1 < 0$ when $a < -1$, the denominator approaches 0. This would cause the overall limit to diverge to infinity, contradicting the given finite limit of $\frac{1}{60}$.

Therefore, we must reject $a = -8.5$. The only valid solution is $a = 7$.

Final Answer:

$$\mathbf{a = 7}$$